

Цилиндры пробоотборные



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Sample Cylinders



Features

- ⦿ Volumes from 40 to 3785 cm (1 gal)
- ⦿ 1/8", 1/4" and 1/2" female NPT connections
- ⦿ Spun cylinder body machined from seamless tubing to provide consistent wall thickness, size and capacity
- ⦿ Cold-formed female NPT threads to provide high strength
- ⦿ All models internally sandblasted to ensure smooth surface without foreign particles
- ⦿ 304L SS and 316L SS available
- ⦿ Cylinders compliant with Transportable Pressure Equipment Directive 2010/35/EU

Application Features

- ⦿ Hydrocarbon sampling for refineries
- ⦿ Gas sampling for chromatography experiments
- ⦿ Condensate sampling for fossil fuel power plants and nuclear power plants
- ⦿ As surge accumulators or reaction vessels
- ⦿ Snubbers in reactor feed lines

Single-Ended Cylinders

- ⦿ 150, 300, and 500 cm³ sizes meet a variety of sampling needs.
- ⦿ 304L stainless steel construction resists intergranular corrosion.

Testing

DOT-4B 500 cylinders are hydrostatically proof tested at 1000 psig (69 bar) minimum.

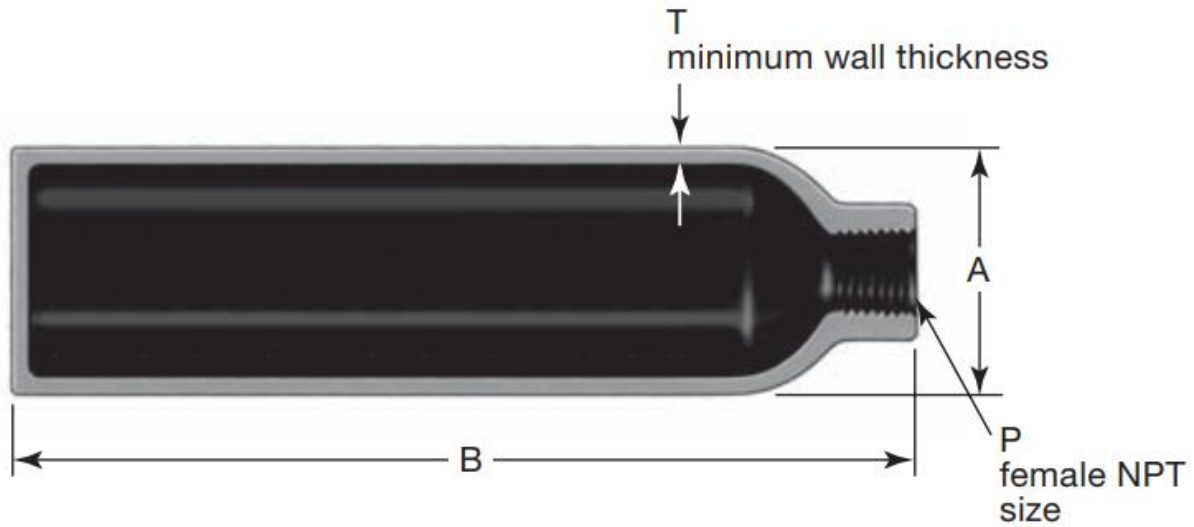
Double-Ended Cylinders

- ⦿ Sizes from 40 to 3785 cm³ (1 gal)
- ⦿ Working pressures up to 5000 psig (344 bar)
- ⦿ 304L and 316L stainless steel materials resist intergranular corrosion.
- ⦿ 304L and 316L stainless steel double-ended cylinders are available with dual certification to DOT and TC requirements.

Testing

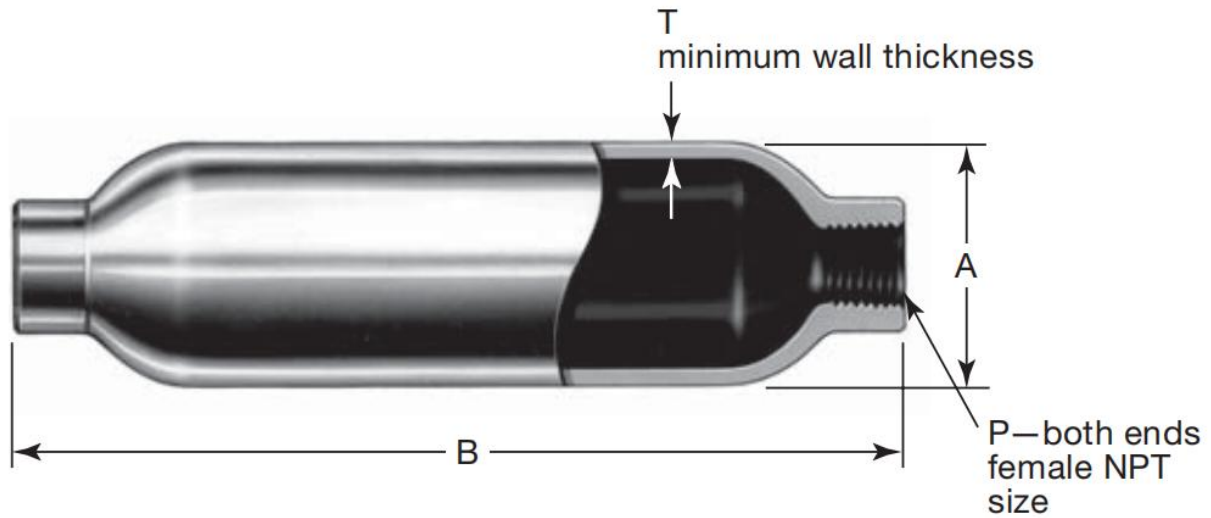
- ⦿ Each DOT cylinder is hydrostatically tested to at least 5/3 the working pressure.
- ⦿ DOT-3E 1800 cylinders are hydrostatically proof tested at 3050 psig (210 bar).
- ⦿ DOT-3A 1800 cylinders are hydrostatically proof tested at 3000 psig (206 bar) minimum.
- ⦿ DOT-3A 5000/ cylinders are hydrostatically proof tested at 8500 psig (586 bar) minimum.

Single-Ended Cylinders



Ordering NO.	Working Pressure Psig (bar)	Internal Volume (cm ³)	P (NPT)	Dimensions in.(mm)			Weight (Kg)
				A	B	T	
-SEC-FN4-0.5K-150	500 (34.5)	150	1/4	2.00 (50.8)	4.88 (124)	0.093 (2.4)	0.5
-SEC-FN4-0.5K-300		300	1/4	2.00 (50.8)	8.62 (219)	0.093 (2.4)	0.82
-SEC-FN4-0.5K-500		500	1/4	2.00 (50.8)	13.6 (345)	0.093 (2.4)	1.2
-SEC-FN4-1.8K-40	1800 (124)	40	1/8	1.25 (31.8)	3.88 (98.6)	0.07 (1.8)	0.14
-SEC-FN4-1.8K-50		50	1/4	1.50 (38.1)	3.75 (95.2)	0.093 (2.4)	0.17
-SEC-FN4-1.8K-75		75	1/4	1.50 (38.1)	4.94 (125)	0.093 (2.4)	0.28
-SEC-FN4-1.8K-150		150	1/4	2.00 (50.8)	5.25 (133)	0.093 (2.4)	0.43
-SEC-FN4-1.8K-300		300	1/4	2.00 (50.8)	8.94 (277)	0.093 (2.4)	0.73
-SEC-FN4-1.8K-400		400	1/4	2.00 (50.8)	11.4 (290)	0.093 (2.4)	0.95
-SEC-FN4-1.8K-500		500	1/4	2.00 (50.8)	13.8 (351)	0.093 (2.4)	1.2
-SEC-FN4-1.8K-1000		1000	1/4	3.50 (88.9)	10.9 (277)	0.18 (4.6)	2.9
-SEC-FN8-1.8K-1000			1/2	3.50 (88.9)	10.9 (277)	0.18 (4.6)	
-SEC-FN4-1.8K-2250	1800 (124)	2250	1/4	4.00 (102)	17.2 (437)	0.21 (5.2)	6.4
-SEC-FN8-1.8K-2250			1/2	4.00 (102)	17.2 (437)	0.21 (5.2)	
-SEC-FN4-1.8K-1gal		3785 (1 gal)	1/4	4.00 (102)	26.7 (678)	0.21 (5.2)	9.5
-SEC-FN8-1.8K-1gal			1/2	4.00 (102)	26.7 (678)	0.21 (5.2)	
-SEC-FN4-5K-150	5000 (345)	150	1/4	1.90 (48.3)	8.00 (203)	0.24 (6.1)	1.4
-SEC-FN4-5K-300		300	1/4	1.90 (48.3)	14.5 (368)	0.24 (6.1)	2.5
-SEC-FN4-5K-500		500	1/4	1.90 (48.3)	23.5 (597)	0.24 (6.1)	4.1

Double-Ended Cylinders



Ordering NO.	Working Pressure Psig (bar)	Internal Volume (cm ³)	P (NPT)	Dimensions in.(mm)			Weight (Kg)
				A	B	T	
-DEC-FN4-0.5K-150	500 (34.5)	150	1/4	2.00 (50.8)	4.88 (124)	0.093 (2.4)	0.5
-DEC-FN4-0.5K-300		300	1/4	2.00 (50.8)	8.62 (219)	0.093 (2.4)	0.82
-DEC-FN4-0.5K-500		500	1/4	2.00 (50.8)	13.6 (345)	0.093 (2.4)	1.2
-DEC-FN4-1.8K-40	1800 (124)	40	1/8	1.25 (31.8)	3.88 (98.6)	0.07 (1.8)	0.14
-DEC-FN4-1.8K-50		50	1/4	1.50 (38.1)	3.75 (95.2)	0.093 (2.4)	0.17
-DEC-FN4-1.8K-75		75	1/4	1.50 (38.1)	4.94 (125)	0.093 (2.4)	0.28
-DEC-FN4-1.8K-150		150	1/4	2.00 (50.8)	5.25 (133)	0.093 (2.4)	0.43
-DEC-FN4-1.8K-300		300	1/4	2.00 (50.8)	8.94 (277)	0.093 (2.4)	0.73
-DEC-FN4-1.8K-400		400	1/4	2.00 (50.8)	11.4 (290)	0.093 (2.4)	0.95
-DEC-FN4-1.8K-500		500	1/4	2.00 (50.8)	13.8 (351)	0.093 (2.4)	1.2
-DEC-FN4-1.8K-1000	1800 (124)	1000	1/4	3.50 (88.9)	10.9 (277)	0.18 (4.6)	2.9
-DEC-FN8-1.8K-1000			1/2	3.50 (88.9)	10.9 (277)	0.18 (4.6)	
-DEC-FN4-1.8K-2250		2250	1/4	4.00 (102)	17.2 (437)	0.21 (5.2)	6.4
-DEC-FN8-1.8K-2250			1/2	4.00 (102)	17.2 (437)	0.21 (5.2)	
-DEC-FN4-1.8K-1gal		3785 (1 gal)	1/4	4.00 (102)	26.7 (678)	0.21 (5.2)	9.5
-DEC-FN8-1.8K-1gal			1/2	4.00 (102)	26.7 (678)	0.21 (5.2)	
-DEC-FN4-5K-150	5000 (345)	150	1/4	1.90 (48.3)	8.00 (203)	0.24 (6.1)	1.4
-DEC-FN4-5K-300		300	1/4	1.90 (48.3)	14.5 (368)	0.24 (6.1)	2.5
-DEC-FN4-5K-500		500	1/4	1.90 (48.3)	23.5 (597)	0.24 (6.1)	4.1

Pressure-Temperature Ratings

Material	SS 316L	SS 316L/304L	SS 304L	Alloy 400	SS 316	SS 304L
Specification	DOT-3A 5000	DOT-3E 1800	DOT-3A 1800	-	-	-
Temperature °C(°F)	Working Pressure, bar (psig)					
-53(-65) to 37 (100)	344 (5000)	124 (1800)	124 (1800)	124 (1800)	69.0 (1000)	34.4 (500)
93 (200)	272 (3960)	93.7 (1360)	93.7 (1360)	108 (1580)	57.8 (840)	34.4 (500)
148 (300)	245 (3570)	84.7 (1230)	84.7 (1230)	102 (1490)	52.3 (760)	34.4 (500)
204 (400)	226 (3290)	77.8 (1130)	77.8 (1130)	98.5 (1430)	48.2 (700)	34.4 (500)
260 (500)	210 (3060)	72.3 (1050)	72.3 (1050)	97.8 (1420)	44.7 (650)	34.4 (500)
315 (600)	201 (2920)	69.0 (1000)	69.0 (1000)	97.8 (1420)	42.7 (620)	34.4 (500)
343 (650)	197 (2870)	67.5 (980)	67.5 (980)	97.8 (1420)	42.0 (610)	34.4 (500)
371 (700)	193 (2810)	66.8 (970)	66.8 (970)	97.8 (1420)	40.6 (590)	34.4 (500)
398 (750)	189 (2750)	65.4 (950)	65.4 (950)	97.1 (1410)	39.9 (580)	34.4 (500)
426 (800)	186 (2700)	64.0 (930)	64.0 (930)	-	39.2 (570)	34.4 (500)
454 (850)	181 (2640)	-	-	-	38.5 (560)	-

Ratings up to 100°F (37°C) are determined by DOT code.

Ratings limited to 300°F (148°C) max with PTFE internal coating.

Ratings may be limited by individual country government regulations.

Options of Internal Surface Treatments

PTFE Coating

The cylinder internal surface can be coated with PTFE to provide nonstick surface for easy cleaning. To order, add -T to the cylinder ordering number. Example: 304L-DEC-FN4-1.8K-300-T.

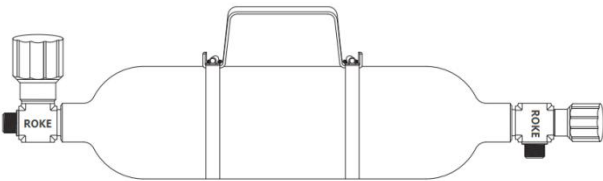
Electropolishing

Electropolishing can provide a clean internal surface with a high degree of passivation. To order, add -EP to the cylinder ordering number. Example: 304L-DEC-FN4-1.8K-300-EP.

Sample Cylinder Accessories

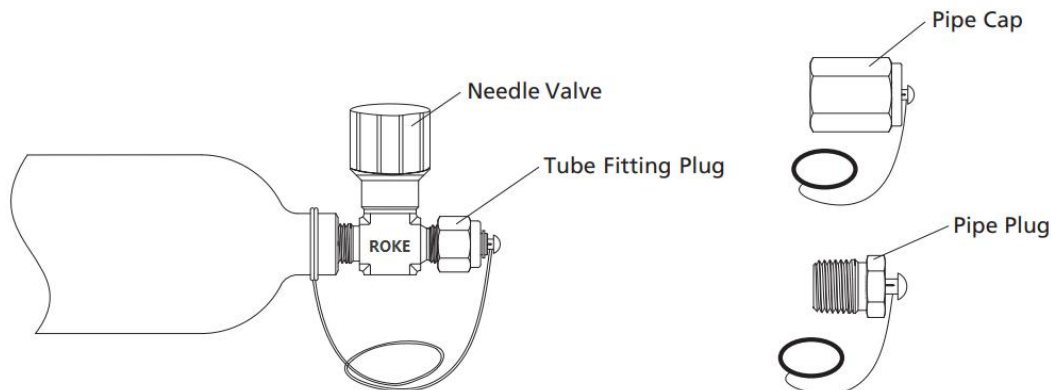
Carrying Handle

The carrying handle provides convenience for transportation of sample cylinders. The handle is made of 304 SS and is available for 300 cm cylinders or larger cylinders.

Carrying Handle	Outside Diameter in. (mm)	Internal Volume cm ³	Basic Ordering No.
	1.9 (48.2)	300~500	HD-CY2-H2
	2.0 (50.8)		
	3.5 (88.9)	1000	HD-CY3-H2
	4.0 (102)	2250/3785	HD-CY4-H4

Caps and Plugs

Caps and plugs are used for cylinder valves to protect the connections (tube fitting or NPT thread) from damages during cylinder transportation.



Outage Tubes

Features

- Made from 316 stainless steel or alloy 400 tubing
- Offered in 1/4 or 1/2 in. NPT sizes

Construction

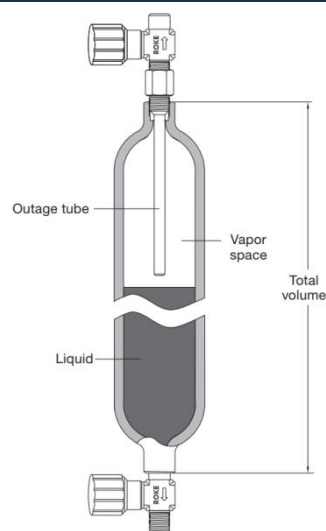
Outage tubes are welded to the male inlet end of an adapter or tee. This adapter or tee is then threaded into the female port of a sample cylinder.

Purpose

Outage tubes provide a vapor space of the desired volume in cylinders containing liquefied gases, so that liquid in the cylinder can expand if the temperature increases. Without enough vapor space, a small temperature increase can cause the liquid to expand and increase the pressure dramatically.

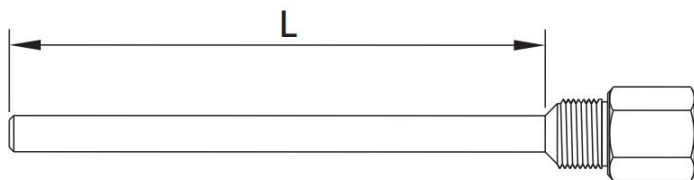
Use

Outage is the vapor space in the cylinder expressed as a percentage of the total volume of the cylinder. % outage = (vapor space/total volume) 100. The cylinder is held vertically with the outage tube at the top, as shown. The length of the outage tube determines the amount of vapor space. Sampling methods and the use of the outage tube are described in technical publications such as ASTM D1265, Standard Practice for Sampling Liquefied Petroleum (LP) Gases (Manual Method).



Outage Tubes Length

Outage tube length (L) is measured from the end of the pipe fitting to the end of the tube.



Tube OD	Volume	Suitable	Minimum Outage, %				
			10	20	30	40	50
			L (Tube Length, in.)				
1/4"	40	Suitable for both ends 1800 psig cylinder	0.87	1.11	1.35	1.59	1.84
5/16"	50		0.85	1.07	1.28	1.50	1.71
	75		1.02	1.34	1.66	1.98	2.31
	150		1.12	1.45	1.79	2.13	2.46
	300		1.65	2.32	2.99	3.67	4.34
	400		2.00	2.90	3.79	4.69	5.59
	500		2.26	3.38	4.50	5.62	6.74
	1000		2.31	3.06	3.81	4.56	5.31
	2250		3.30	4.59	5.88	7.17	8.46
	3785 (1 gal)		4.62	6.79	8.96	11.14	13.31
1/2"	1000		2.21	2.96	3.71	4.46	5.21
	2250		3.30	4.59	5.88	7.17	8.46
	3785 (1 gal)		4.52	6.69	8.86	11.04	13.21
5/16"	150	Suitable for single end 500 psig cylinder	1.09	1.43	1.77	2.12	2.46
	300		1.59	2.27	2.96	3.65	4.34
	500		2.16	3.30	4.45	5.60	6.74
	150	Suitable for both ends 5000 psig cylinder	1.62	2.17	2.71	3.26	3.81
	300		2.74	3.84	4.93	6.03	7.12
	500		4.39	6.21	8.04	9.86	11.68

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